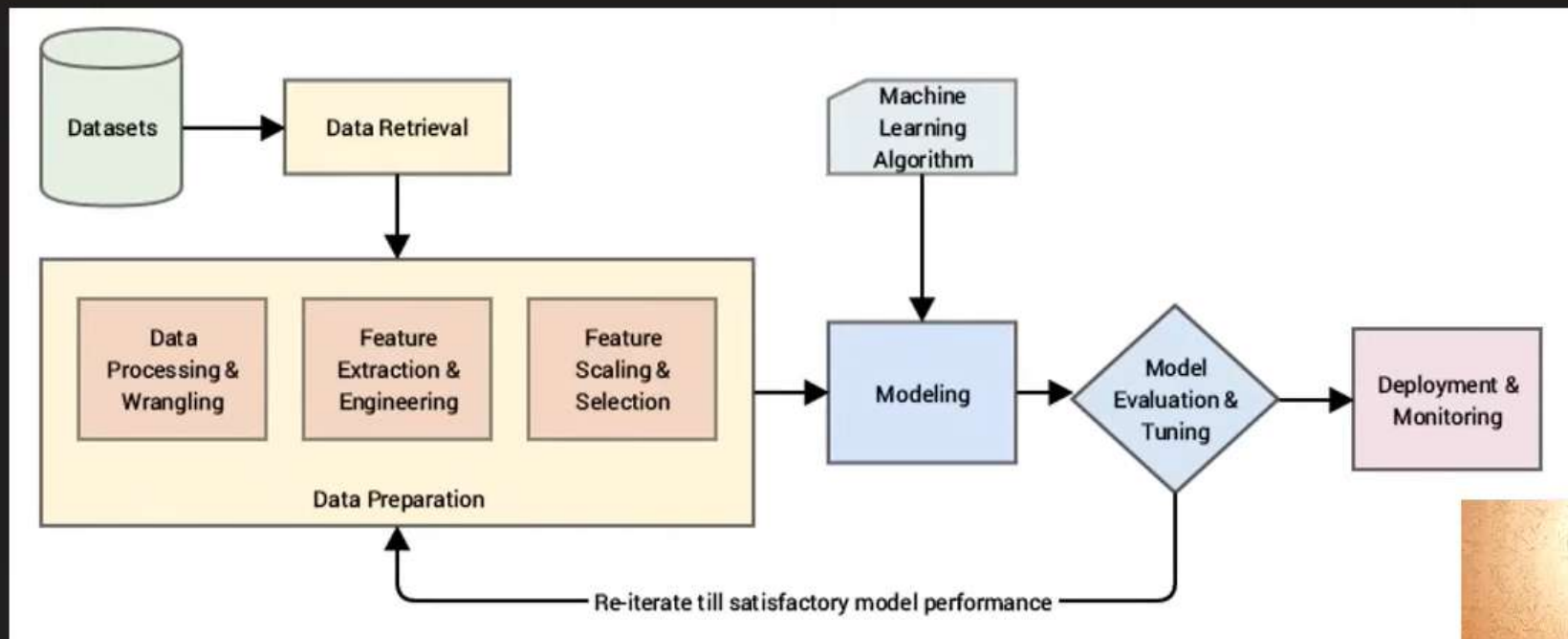


Feature Engineering

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I

Feature engineering is the process of using domain knowledge to extract features from raw data. These features can be used to improve the performance of machine learning algorithms.



Feature

Feature Engineering

Feature Transformation

Feature Construction



Feature Selection

Feature Extraction

- Missing Value Imputation
- Handling Categorical Features
- Outlier Detection
- Feature Scaling

Classifying feature Engineering



1.1 Missing Values Imputation

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Feature Engineering

1.1 Missing Values Imputation

1.2 Handling Categorical Valu...

1.3 Outlier Detection

1.4 Feature Scaling

2. Feature Construction

3. Feature Selection

4. Feature Extraction



Average_Age = 26.0

ID	City	Age	Married ?
1	Lisbon	25	0
2	Berlin	25	1
3	Lisbon	30	1
4	Lisbon	30	1
5	Berlin	18	0
6	Lisbon	NaN	0
7	Berlin	30	1
8	Berlin	NaN	0
9	Berlin	25	1
10	Madrid	25	1



ID	City	Age	Married ?
1	Lisbon	25	0
2	Berlin	25	1
3	Lisbon	30	1
4	Lisbon	30	1
5	Berlin	18	0
6	Lisbon	26	0
7	Berlin	30	1
8	Berlin	26	0
9	Berlin	25	1
10	Madrid	25	1

1.2 Handling Categorical Values

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Index	Animal
0	Dog
1	Cat
2	Sheep
3	Horse
4	Lion

One-Hot code



Index	Dog	Cat	Sheep	Lion	Horse
0	1	0	0	0	0
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	0	1
4	0	0	0	1	0



1.3 Outlier Detection

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Feature Engineering

1.1 Missing Values Imputation

1.2 Handling Categorical Val...

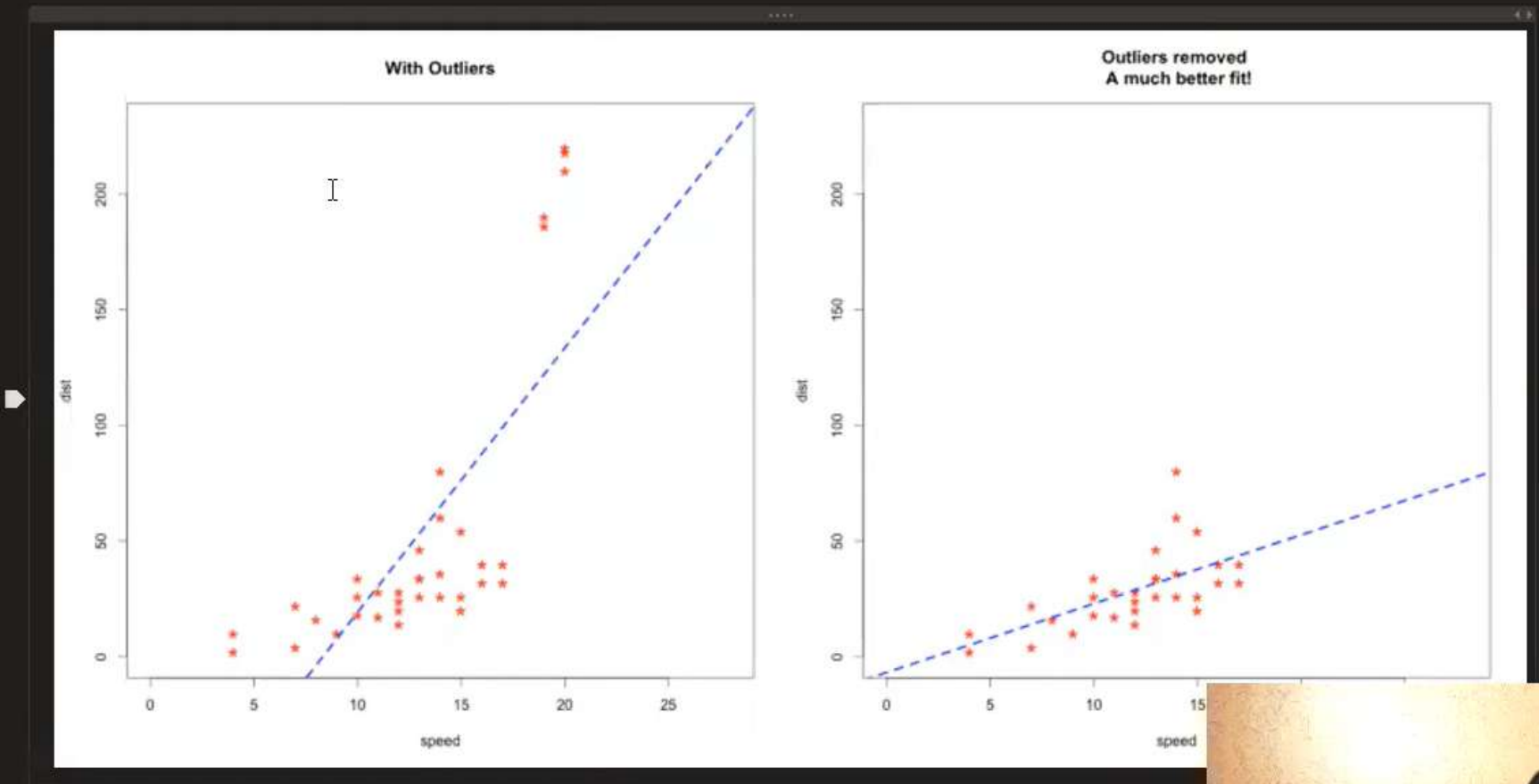
1.3 Outlier Detection

1.4 Feature Scaling

2. Feature Construction

3. Feature Selection

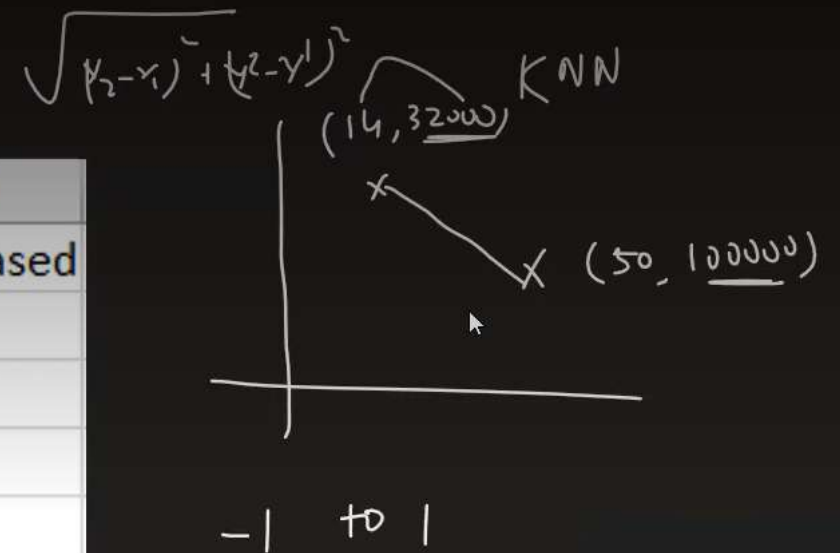
4. Feature Extraction



1.4 Feature Scaling

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	A	B	C	D
1	Country	Age	Salary	Purchased
2	France	44	72000	No
3	Spain	27	48000	Yes
4	Germany	30	54000	No
5	Spain	38	61000	No
6	Germany	40		Yes
7	France	35	58000	Yes
8	Spain		52000	No
9	France	48	79000	Yes
10	Germany	50	83000	No
11	France	37	67000	Yes



2. Feature Construction

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P. Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	-	1	0	A/5 21171	7.25	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs)	female	38	1	-	0	PC 17599	71.2833	C85
3	1	3	Heikkinen, Miss. Laina	female	26	0	-	0	STON/O2.	7.925	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	-	1	0	113803	53.1	C123
5	0	3	Allen, Mr. William Henry	male	35	0	0	0	373450	8.05	S
6	0	3	Moran, Mr. James	male		0	0	0	330877	8.4583	Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	0	17463	51.8625	E46
8	0	3	Palsson, Master. Gosta Leonard	male	2	-	3	1	349909	21.075	S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmin)	female	27	0	2	2	347742	11.1333	S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	0	237736	30.0708	C

family

$\left\{ \begin{array}{l} 0 - \text{Alone} \\ 0-4 - \text{small} \\ 4 < - \text{large} \end{array} \right\}$



3. Feature Selection

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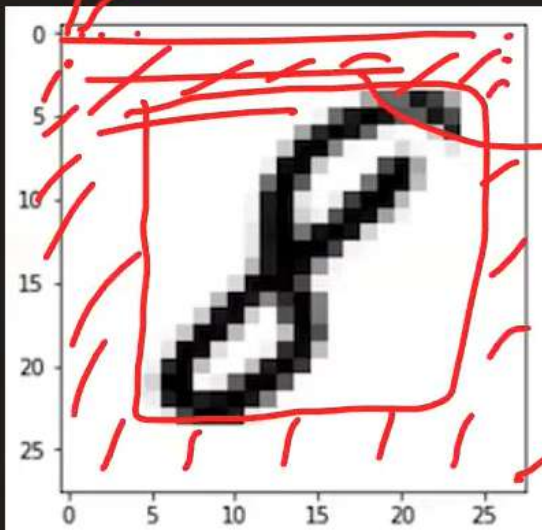
```
In [5]: train_df = pd.read_csv(f'{PATH}train.csv', header = 0)
```

```
In [19]: train_df
```

```
Out[19]:
```

	label	pixel0	pixel1	pixel2	pixel3	pixel4	pixel5	pixel6	pixel7	pixel8	...	pixel774
0	1	0	0	0	0	0	0	0	0	0	...	0
1	0	0	0	0	0	0	0	0	0	0	...	0
2	1	0	0	0	0	0	0	0	0	0	...	0
3	4	0	0	0	0	0	0	0	0	0	...	0

784 features



MNIST

50000
imgs

$$28 \times 28 = 784$$



4. Feature Extraction ✓

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